

GharVal AI

Technical Documentation & Machine Learning Architecture

Target Market:	Nepal (Kathmandu Valley / Bhaktapur)
Platform:	GharVal AI (■■■-■■■■■ ■■■■■■■■■)
Author:	Lead Data Scientist
Algorithms:	Random Forest (Champion) & XGBoost
Status:	Active & Verified on Bhaktapur Listings

1. System Architecture

The project is structured as a modular, decoupled Machine Learning pipeline that cleanly separates data generation, exploratory analysis, training, and the final application interface. Each component runs as an independent stage of the data engineering cycle:

Component	Filename / Path	Function / Output
Data Engine	src/generate_data.py	Synthesizes correlated housing CSV data
EDA Engine	src/eda.py	Exports statistical analysis heatmaps/plots
Training Pipeline	src/train.py	Trains & evaluates models, exports model pickles
User Interface	app.py	Interactive Nepalese valuation web app

2. Data Strategy & Dataset Architecture

To deliver high-fidelity estimations, the project utilizes a high-quality simulated house valuation dataset containing 1,000 observations. Programmatic synthesis allows structural correlation to remain perfectly consistent and ensures absolute reproducibility. The house price is generated using a realistic structural formula incorporating non-linear quality multipliers and randomized Gaussian noise:

$$\text{Price (USD)} = \$40,000 + (\$115 * \text{TotalSqFt}) + (\$22,000 * \text{OverallQuality}) + (\$3.5 * \text{TotalSqFt} * \text{OverallQuality}) + (\$18,000 * \text{Bathrooms}) + (\$10,000 * \text{Bedrooms}) + (\$950 * \text{DeltaYear}) + \text{Noise}$$

3. Feature Definitions & Nepalese Localization

To customize the application for standard buyers and developers in Nepal, local engineering units and parameters have been integrated directly into the inference layer:

Feature	Unit / Scale	Nepalese Market Context
Area (TotalSqFt)	SqFt (500 - 5000)	Converted dynamically to Anna measurements (1 Anna ≈ 342.25 SqFt)
Overall Quality	Discrete (1 - 10)	Correlates structure: 1-3 masonry, 4-7 RCC frame, 8-10 high-end luxury villa
Year Built	Year (1950 - 2026)	Evaluates seismic safety premium for post- 2015 Gorkha Earthquake construction
Road Width	Feet (10 - 26)	Access road width in feet (mostly 13ft blacktopped or 20ft RCC)
Road Type RCC	Binary (0 or 1)	1 if access road is concrete (RCC) pavement, 0 if blacktopped

Estimated Price	NPR (₹. / Rupees)	Scaled via exchange index (135x) and formatted into Lakhs/Crores standard
-----------------	-------------------	---

4. Model Comparisons & Training Mechanics

The preprocessing layer splits the dataset into 80% train and 20% test sets, scaling inputs via StandardScaler. The pipeline trains two distinct ensemble algorithms: **Random Forest Regressor** and an **XGBoost (Extreme Gradient Boosting) Regressor**. The evaluations show exceptional convergence:

Model	Test RMSE	Test R ² Score	Verdict
Random Forest	\$17,890.62	0.8317	Champion (Uncorrelated noise reduction)
XGBoost Regressor	\$18,634.74	0.8174	Excellent performance, slight variance loss

Why Random Forest Performed Best

The Bhaktapur housing dataset incorporates randomized land price indices per Anna to simulate localized market pricing variations. Random Forest's bootstrap aggregation (bagging) is mathematically designed to minimize prediction variance by averaging 100 independent decision trees, smoothing out these uncorrelated land price noise factors more effectively than XGBoost's sequential boosting updates.

5. Interactive Inference Layer

The Streamlit dashboard (app.py) provides real-time model evaluation:

- 1. Preprocessing Pipeline:** Inputs entered via sliders are immediately structured into a single-row dataframe, normalized via the pre-fit scaler object (scaler.pkl).
- 2. Model Prediction:** The champion Random Forest model (best_model.pkl) calculates the value.
- 3. Unit Conversions:** Area is mapped to Anna; valuation is converted to Nepalese Rupees and represented in Lakhs / Crores notation.
- 4. Seismic Adjustments:** Structural pricing deductions are applied to houses built pre-2015 (Kathmandu earthquake standards), prioritizing building code compliance.

GharVal AI System Validation

Status: Verified

Local Web App: <http://localhost:8501>